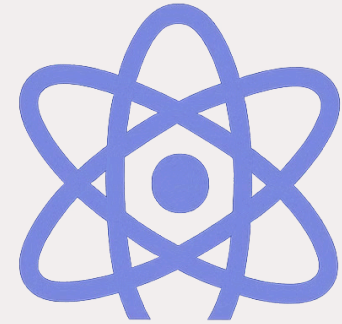
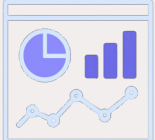


Proof of Concept: Discourse Network Analysis

Indonesia's Nuclear Policy Debate - RUU EBET





Executive Summary



Integrated Methodology

Combining Discourse Network Analysis, Social Network Analysis, and AI-powered sentiment classification



Strategic Objective

Automatically map policy actors, positions, and coalitions from news text with measurable accuracy



Deliverables

Interactive network visualizations, sentiment dashboards, and coalition matrices for stakeholder presentation

Core Analytical Framework

Three-Pillar Approach

This PoC integrates complementary analytical methods to transform raw policy discourse into strategic intelligence.



Discourse Network Analysis

Maps actors, positions, and arguments in policy debates using structured codebooks and AI extraction



Social Network Analysis

Quantifies coalition formation, brokerage roles, and conflict patterns through network metrics



Sentiment Analysis

Classifies textual tone using IndoBERT, distinguishing article stance from actor positions

Data Sources & Pipeline Architecture

01

CSV News Articles

Articles from national news portals with keywords, categories, and content

03

Classification

IndoBERT for sentiment (positive/negative/neutral) and Mistral AI for DNA coding

05

Network Generation

SNA metrics (degree, betweenness, eigenvector centrality)

02

Preprocessing & Cleaning

Null removal, text normalization, and content filtering for analysis readiness

04

Matrix Construction

Affiliation (which actors share the same concepts), congruence (how aligned two actors are), and conflict matrices (how opposed two actors are) from coded actor-concept relationships

06

Report Export

Documentation, Report, and interactive visualization

Key Concepts & Definitions

Actors & Coalitions

Individuals, institutions, or groups expressing positions in nuclear policy debates. Categorized by type (Legislative, Executive, BUMN, Academia, Civil Society, International) and stance (PRO, ANTI, MODERAT, REGULATOR).

Discourse Concepts

Propositional claims about nuclear policy that actors support (+1) or oppose (-1). Examples: "Nuclear power suitable for national energy mix" or "SMR technology safe for Indonesia."



PRO Coalition

Support development



ANTI Coalition

Reject due to risks



MODERAT Coalition

Support with strict conditions



REGULATOR Coalition

Neutral oversight: BAPETEN

AI Models & Methodology

IndoBERT Sentiment Classifier

Model: w11wo/indonesian-roberta-base-sentiment-classifier

Classifies article tone as positive, negative, or neutral with confidence scores

Mistral AI Coding Engine

Model: mistral-medium-latest API

Extracts actors, concepts, and stances from news text using structured codebooks

Network Metrics

- Degree centrality: coalition breadth ("How many connections?")
- Betweenness centrality: brokerage power ("How often do you act as a bridge?")
- Eigenvector centrality: influence through connections ("How influential are your connections?")

DNA Components

- Affiliation (which actors share the same concepts),
- Congruence (how aligned two actors are),
- Conflict matrices (how opposed two actors are)



Identifying Organic vs. Non-Organic Accounts (Buzzer Detection)

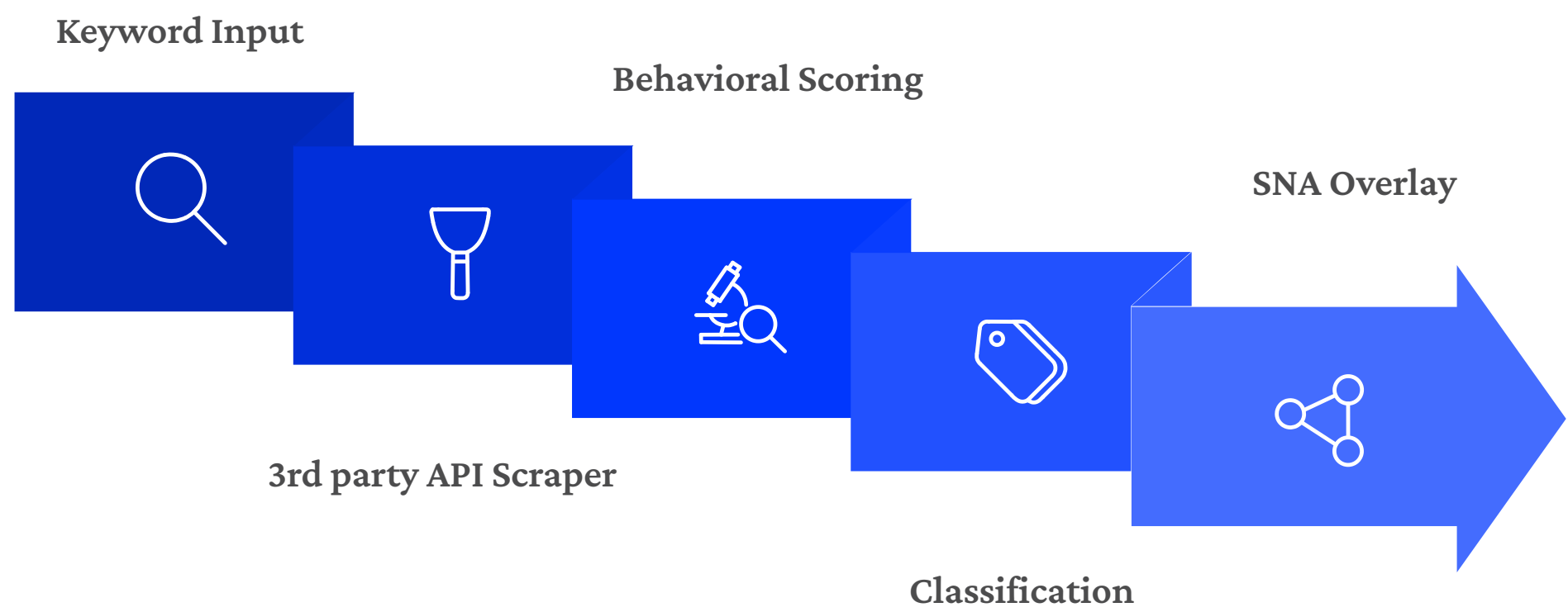
Organic Accounts

- Genuine individual users
- Varied posting patterns
- Natural language variation

Non-Organic (Buzzer)

- Coordinated amplification
- High-frequency repetitive posting
- Abnormal followers/following ratio

Proposed Data Pipeline

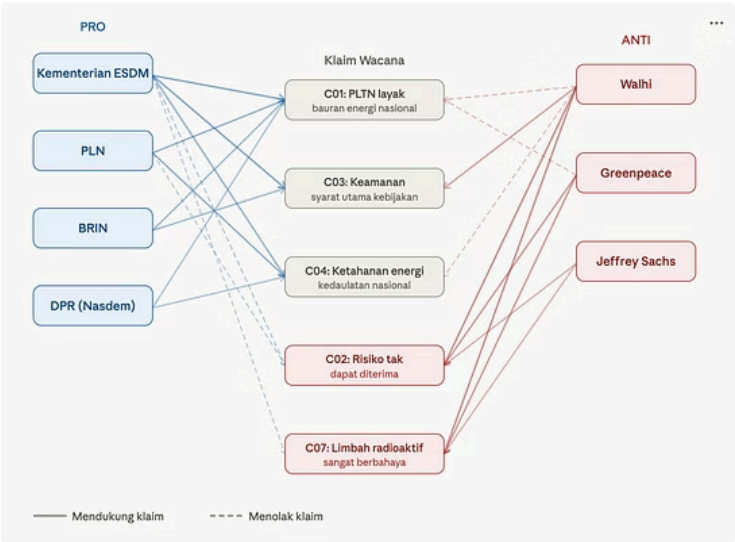
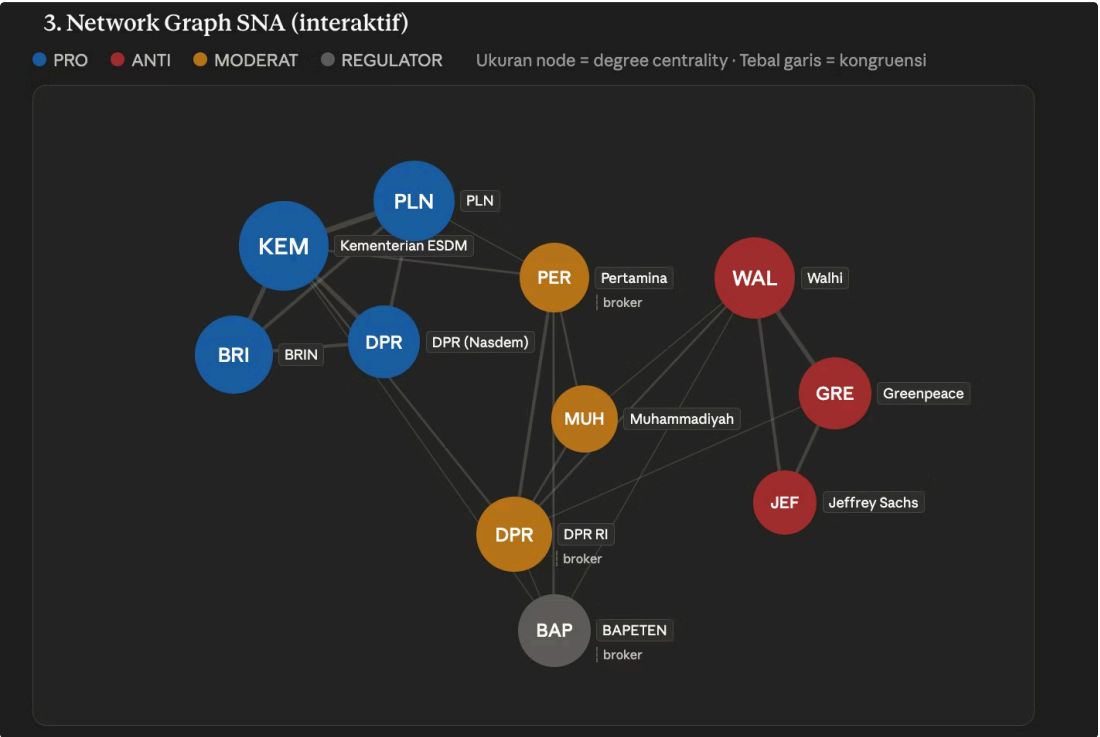
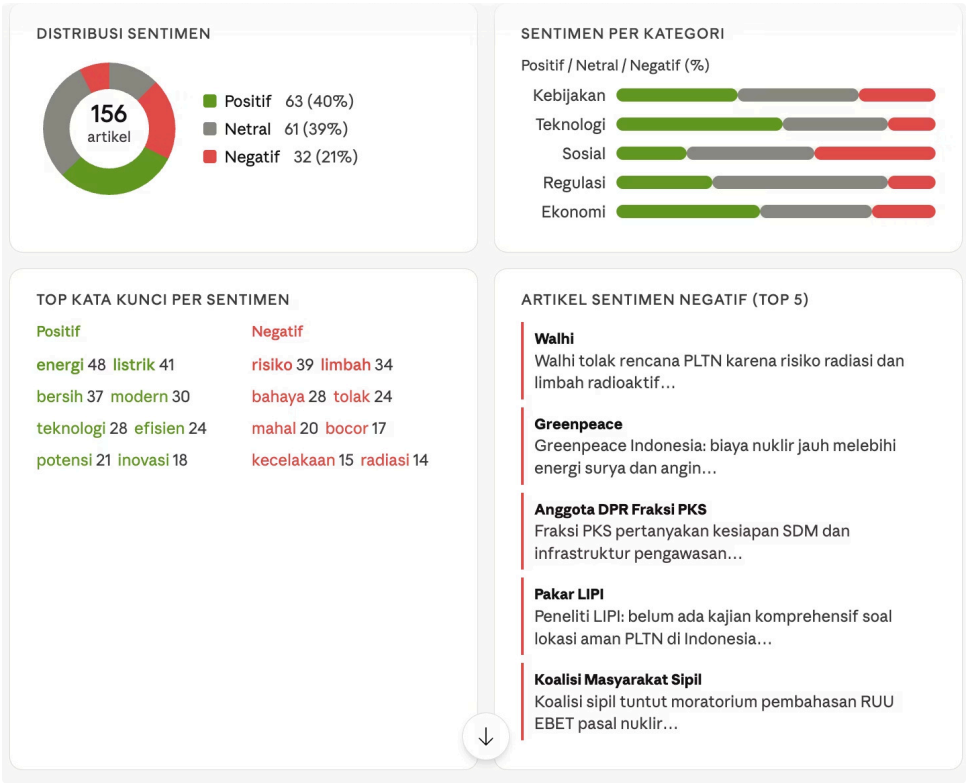


Possible Scoring Tools — Under Evaluation

Botometer Per-username bot score (⚠️ API access limited)
Manual indicators Timestamp, ratio, content patterns (✓ Always feasible)
Twibot / others Alternative bot classifiers (⚠️ To be evaluated)

Note: Final tool selection depends on platform API access, data availability, and tools. Manual behavioral indicators will serve as fallback scoring method.

Visualizations & Deliverables



Current Limitations & Constraints

Data Scope Constraints

Only online news articles analyzed (150+ rows). Missing social media (Twitter/X, Instagram), video transcripts, and English-language sources limit comprehensiveness.

Codebook Completeness

Manual actor and concept dictionaries may miss emerging actors or arguments. Articles without identified actors are automatically excluded from DNA analysis.

Manual Keyword Input & Data Volume Definition

Manual keyword dictionaries must be carefully defined upfront for scraping to work effectively.

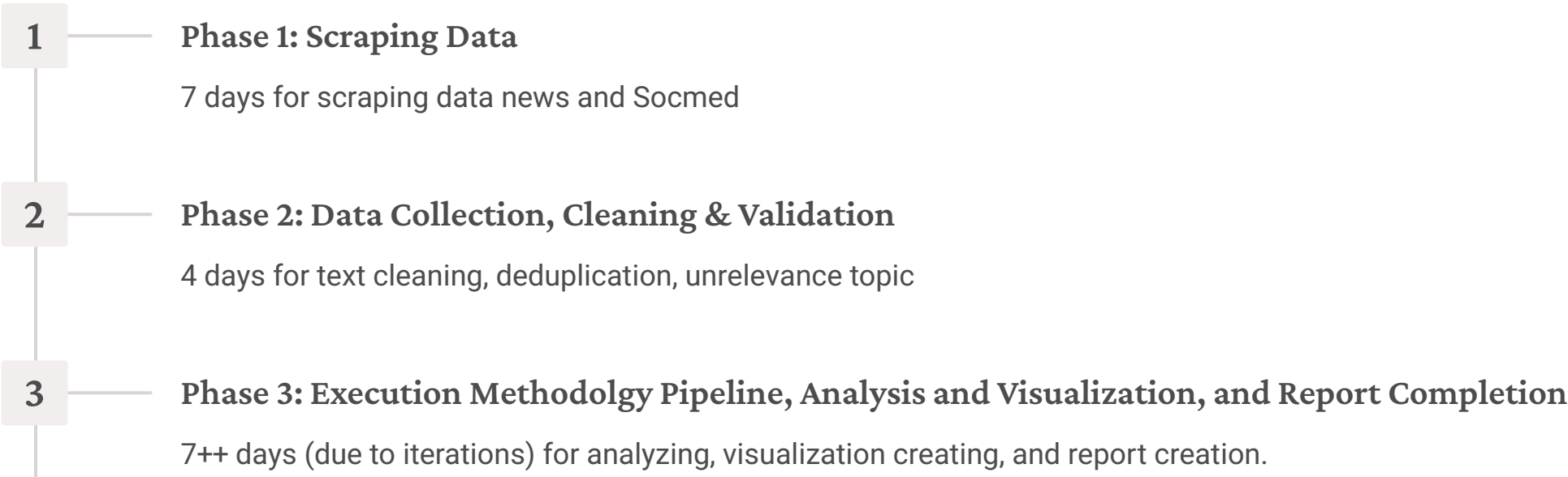
The scope of articles to collect needs to be predetermined to balance comprehensiveness with analysis capacity. Without clear keyword parameters and volume targets, scraping may return irrelevant results or insufficient data for robust network analysis.



Development Roadmap

3-Week Timeline Overview

A 3-week plan to expand data collection, strengthen validation, and complete the analysis workflow for this one-time project.



Scraping Methodology

- Use Diffbot API for structured news article extraction
- Use Apify for social media crawling

Specific Data Sources

News coverage from online portals, plus social media posts or comments from Socmed with hashtags such as #RUUEbet and #NuklirIndonesia or other keywords.

Extraction Methods

Apply keyword-based collection for news, API-driven extraction for article metadata and text, and crawler-based harvesting for social media content to support broader source coverage.

Next Steps:

Expected Outcomes

- List of data news with links
- A completed one-time analysis project with validated findings
- comprehensive network insights and its visualization

Questions:

1. Do you provide lists of keywords for crawling? or just fetch keyword from "7 Kategori & Variable"?
2. Source Socmed (FB, Tiktok, YT, IG, X)?
3. How long the data period?
4. Should we just focus on DNA or better include SNA and sentiment as well?
5. Do you have another expected output (visualizations or else)?
6. Do we still want to detect organic and non-organic (buzzer)?